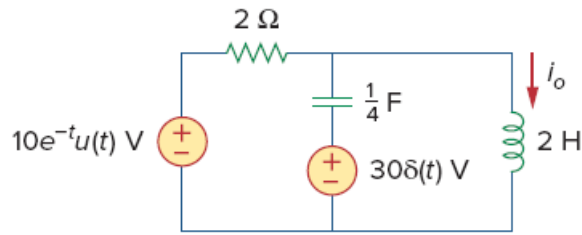


Problem

Determine the Fourier transform of $i_o(t)$ in the circuit of Fig. 18.48.

Figure 18.48



Step-by-step solution

Step 1 of 3

(Refer to Figure 18.48 in the textbook) represent the circuit in the frequency domain.

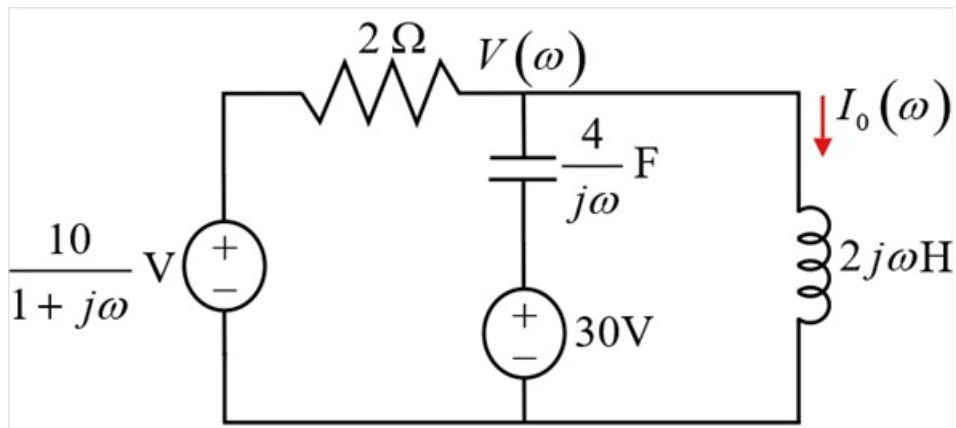


Figure 1

Step 2 of 3

Apply the nodal analysis in the Figure 1.

$$\frac{V(\omega) - \left(\frac{10}{1+j\omega}\right)}{2} + \frac{V(\omega) - 30}{\left(\frac{4}{j\omega}\right)} + \frac{V(\omega)}{2j\omega} = 0$$

$$V(\omega) \left[\frac{1}{2} + \frac{j\omega}{4} - \frac{j}{2\omega} \right] - \frac{10}{2(1+j\omega)} - \frac{30j\omega}{4} = 0$$

$$V(\omega) \left[\frac{2\omega + j\omega^2 - j2}{4\omega} \right] = \frac{20 + 30j\omega(1+j\omega)}{4(1+j\omega)}$$

$$V(\omega) = \frac{20\omega + 30j\omega^2(1+j\omega)}{(1+j\omega)(2\omega + j\omega^2 - j2)}$$

Step 3 of 3

The current $I_0(\omega)$ across the inductor is,

$$I_0(\omega) = \frac{V(\omega)}{2j\omega}$$

Substitute $\frac{20\omega + 30j\omega^2(1+j\omega)}{(1+j\omega)(2\omega + j\omega^2 - j2)}$ for $V(\omega)$.

$$\begin{aligned} I_0(\omega) &= \frac{\left[\frac{20\omega + 30j\omega^2(1+j\omega)}{(1+j\omega)(2\omega + j\omega^2 - j2)} \right]}{2j\omega} \\ &= \frac{10 + 15j\omega(1+j\omega)}{(1+j\omega)(2j\omega - \omega^2 + 2)} \\ &= \frac{10 + 15j\omega(1+j\omega)}{(1+j\omega)(j\omega + 1)^2} \end{aligned}$$

Hence, the current across the inductor is $\boxed{\frac{10 + 15j\omega(1+j\omega)}{(1+j\omega)(j\omega + 1)^2} \text{ A}}$.